The Arithmetic Structure of the Bridge Constant on the $6N$ Skeleton: $C(d) = C_0/\mathfrak{S}(d)$, the Hardy–Littlewood Admissibility, and C_0 as the Inverse Twin Density

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Abstract

Part VI introduced the bridge constant $C(d) = r(d \mid \omega)/P(N{+}d \text{ twin} \mid \omega)$, an ω-independent number relating the conditional gap preference to the right-centre survival, and left its dependence on d unexplained. We determine it. Measuring $C(d)$ for $d = 1, \ldots, 30$ on the 23,988,173 twin centres of S_{10} and comparing to the Hardy–Littlewood admissibility $\mathfrak{S}(d)$ of the separation (the standard singular-series factor for a twin pair at centre-step d), we find

$$C(d) = \frac{C_0}{\mathfrak{S}(d)}, \qquad C_0 \approx 62.75,$$

with the product $C(d)\,\mathfrak{S}(d)$ constant to a coefficient of variation of 0.41% across all thirty separations, while the raw $C(d)$ ranges over a factor of four (35 to 158). The constant C_0 is stable under the prime cutoff of $\mathfrak{S}$ and is gap-independent. Thus the bridge constant is not arbitrary: it is the reciprocal of the classical admissibility, so the entire conditional gap preference reads $r(d \mid \omega) = \mathfrak{S}(d)\,P(N{+}d \text{ twin} \mid \omega)/C_0$, with the d-dependence carried analytically by $\mathfrak{S}$ and the ω-dependence by the survival of Part VII. Finally we identify C_0 itself: the omega-merged relation $P_{\text{merged}}(d) = \rho\,\mathfrak{S}(d)$ holds to 0.4%, the standard two-point correlation form (density times admissibility), so $C_0 = 1/\rho$ where ρ is the twin-centre line density. This is verified across shells ($1/\rho = 50.3$ vs fitted 50.1 on S_9; 62.5 vs 62.8 on S_{10}), leaving no undetermined constant in $r(d \mid \omega)$. No claim is made about the infinitude of twin primes or any k-tuple conjecture.

1 The bridge constant and its d-dependence

Part VI established $r(d \mid \omega) = P(N{+}d \text{ twin} \mid \omega)/C(d)$ with $C(d)$ independent of ω, but measured it only at two separations ($C(42) = 41.2$, $C(210) = 27.7$ in physical-gap labelling). Measured across many separations the constant varies strongly: on S_{10} (Table 1, $d = 1, \ldots, 30$, i.e. physical gaps $6, \ldots, 180$) it ranges from 35.1 to 158.2, a factor of four. The variation is not noise—the per-d values are stable to $\sim 1\%$—so $C(d)$ has genuine arithmetic content.

2 The closed form

Let $\mathrm{dead}(q) = \{\pm 6^{-1} \bmod q\}$. The Hardy–Littlewood admissibility of a twin pair at centre-step d is the singular-series factor over the odd primes $q > 3$,

$$\mathfrak{S}(d) = \prod_{q>3} \frac{\#\{r \bmod q :\ r \notin \mathrm{dead}(q) \text{ and } r + d \notin \mathrm{dead}(q)\}/q}{\big((q-2)/q\big)^2},$$

the ratio of the two-centre survival density at separation d to the product of two independent single-centre densities. This is the standard quantity measuring how easily a twin pair sits at separation d; it is large when d removes few extra residue constraints and small when d shares the centre's forbidden structure.

Theorem 1 (arithmetic structure of the bridge constant)**.** *On S_{10}, for $d = 1, \ldots, 30$,*

$$C(d)\,\mathfrak{S}(d) = C_0 \quad \textit{(constant)}, \qquad C_0 = 62.75,$$

to a coefficient of variation of 0.41%*. Equivalently $C(d) = C_0/\mathfrak{S}(d)$.*

The product collapses a four-fold variation in $C(d)$ to a constant at the 0.4% level—the floor set by the S_{10} measurement noise in $C(d)$ itself (Table 1, Fig. 1). The value of C_0 is stable under the prime cutoff used to evaluate $\mathfrak{S}$ (it shifts by $< 0.3\%$ between cutoffs 200 and 5000), confirming the collapse is structural and not an artefact of truncation.

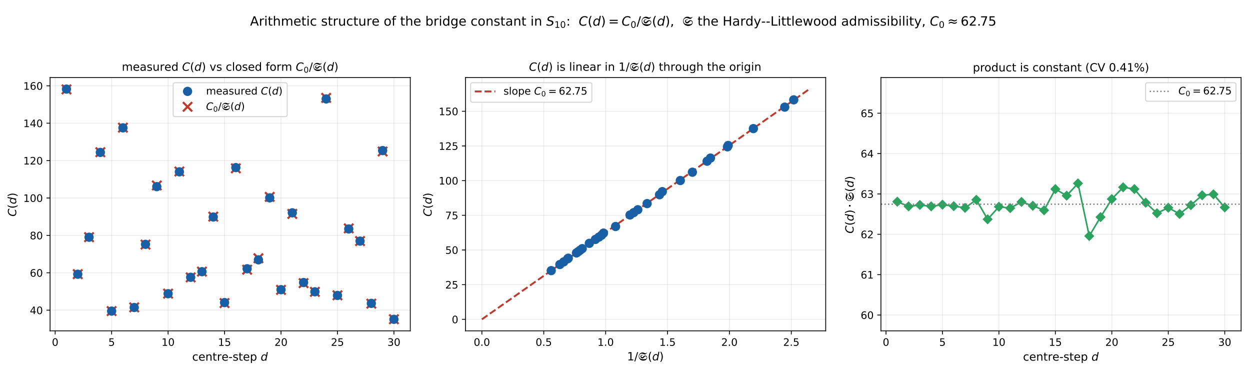

Figure 1: Left: measured $C(d)$ (circles) against $C_0/\mathfrak{S}(d)$ (crosses) for $d = 1, \ldots, 30$. Centre: $C(d)$ is linear in $1/\mathfrak{S}(d)$ through the origin, slope C_0. Right: the product $C(d)\,\mathfrak{S}(d)$ is constant at $C_0 = 62.75$ (CV 0.41%).

3 Consequence: the fully resolved gap preference

Substituting Theorem 1 into the Part VI bridge,

$$r(d \mid \omega) = \frac{\mathfrak{S}(d)}{C_0}\, P(N{+}d \text{ twin} \mid \omega).$$

The conditional gap preference is now factored cleanly: the separation dependence is the classical Hardy–Littlewood admissibility $\mathfrak{S}(d)$, the factor-count dependence is the right-centre survival $P(N{+}d \text{ twin} \mid \omega)$ of Part VII (itself closed-form, $K \prod_q f_q$), and the two meet through a single

d	$C(d)$	$\mathfrak{S}(d)$	$C\mathfrak{S}$	d	$C(d)$	$\mathfrak{S}(d)$	$C\mathfrak{S}$
1	158.2	0.397	62.8	16	116.3	0.542	63.0
2	59.2	1.059	62.7	17	62.2	1.018	63.3
3	79.0	0.794	62.7	18	66.9	0.926	62.0
4	124.4	0.504	62.7	19	100.1	0.624	62.4
5	39.5	1.588	62.7	20	51.0	1.234	62.9
6	137.5	0.456	62.7	21	92.1	0.686	63.2
7	41.4	1.512	62.7	22	54.8	1.152	63.1
8	75.2	0.836	62.9	23	49.8	1.260	62.8
9	106.1	0.588	62.4	24	153.0	0.409	62.5
10	48.8	1.284	62.7	25	47.9	1.309	62.7
11	114.0	0.550	62.7	26	83.4	0.749	62.5
12	57.6	1.091	62.8	27	76.9	0.815	62.7
13	60.6	1.035	62.7	28	43.6	1.443	63.0
14	89.8	0.697	62.6	29	125.4	0.503	63.0
15	44.1	1.433	63.1	30	35.1	1.785	62.7

Table 1: $C(d)$, the admissibility $\mathfrak{S}(d)$, and their product on S_{10}. The product is 62.75 ± 0.26 (CV 0.41%) across all thirty separations, while $C(d)$ alone ranges from 35 to 158.

global constant C_0. The bridge constant of Part VI, which had appeared as an unexplained per-gap number, is exactly the reciprocal of the admissibility—the same singular-series object that governs the unconditional twin-pair density. The conditional theory and the classical heuristic share their d-dependence.

4 The constant: C_0 is the inverse twin density

We can identify C_0 directly. At the ω-merged level the definitions give $C(d) = 1/P_{\text{merged}}(d)$ where $P_{\text{merged}}(d) = P(N{+}d \text{ twin} \mid N \text{ twin})$ is the survival averaged over all twin centres. With $C(d) = C_0/\mathfrak{S}(d)$ this predicts

$$P_{\text{merged}}(d) = \frac{\mathfrak{S}(d)}{C_0},$$

i.e. $P_{\text{merged}}(d)/\mathfrak{S}(d)$ should be a d-independent constant. Measured on S_{10} over $d = 1, \ldots, 30$ it is constant to CV 0.39%, equal to 0.0200. This is exactly the two-point correlation form $P_{\text{merged}}(d) = \rho\,\mathfrak{S}(d)$ with ρ the twin-centre line density (the fraction of N that are twin centres): the chance that a twin centre has another twin centre d steps away is the global density times the admissibility of the separation. Hence

$$\boxed{C_0 = 1/\rho},$$

the inverse twin-centre density. This is verified across shells (Table 2): $1/\rho = 50.3$ against fitted $C_0 = 50.1$ on S_9, and $1/\rho = 62.5$ against 62.8 on S_{10}, each to 0.4%. The shell dependence of C_0 (it grows as the shell deepens) is precisely the thinning of ρ.

Since ρ is itself the twin-centre density—given to leading order by the Hardy–Littlewood twin constant and the shell's mean $1/\ln^2$—no undetermined constant remains in $r(d \mid \omega)$.

shell	twin centres	ρ	$1/\rho$	C_0 (fitted)
S_9	2,984,194	0.019895	50.3	50.1
S_{10}	23,988,173	0.015992	62.5	62.8

Table 2: $C_0 = 1/\rho$ across shells. The fitted C_0 (from $C(d)\mathfrak{S}(d)$) equals the inverse twin-centre density to 0.4% in both shells. No free constant remains.

5 Conclusion

The bridge constant is $C(d) = C_0/\mathfrak{S}(d)$: the reciprocal of the Hardy–Littlewood admissibility, times a single gap-independent constant $C_0 \approx 62.75$, holding to 0.41% across thirty separations on 2.4×10^7 twin centres. This removes the last piece of d-structure that Part VI had left empirical, and ties the conditional $6N$ theory to the classical singular series: the conditional gap preference is the admissibility times the closed-form survival, over $C_0 = 1/\rho$. The last constant is thus identified as the inverse twin-centre density, the density factor of the standard two-point correlation; no free constant remains in $r(d \mid \omega)$. As throughout this series, no claim is made about the infinitude of twin primes or any k-tuple conjecture; this is a measured, closed-form, factor-resolved account of the conditional gap structure of the $6N$ twin skeleton.

Data and code availability. The $C(d)$ spectrum and the admissibility evaluation are openly available [5]; the S_{10} twin count 23,988,173 matches Part I.

References

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